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(See 6.13)

# MILITARY SPECIFICATION

## CONTACTORS AND CONTROLLERS, ELECTRIC MOTOR AC OR DC, AND ASSOCIATED SWITCHING DEVICES

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers alternating current (ac) and direct current (dc) motor controllers and associated switching devices. Included are manual and magnetic controllers, controller parts, limit switches, pressure switches, temperature switches, selector switches, push-buttons, and similar devices used for control of electric motors. Semiconductor (solid-state) devices used as parts for magnetic controllers are included.

1.2 Classification.

1.2.1 Controller. Controllers shall be designated in the following form and as specified (see 6.2 and 6.11).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.1 Power. The power shall be designated by the corresponding one letter symbol and is limited to the following:

AC - Alternating current
DC - Direct current

1.2.1.2 Type. Controllers shall be designated by the corresponding two letter symbols as follows:

MN - Manual
MG - Magnetic

1.2.1.3 Configuration. Configuration shall be designated by the corresponding letter symbols as follows:

AL - Across-the-line
AU - Reduced voltage, autotransformer
SI - Reduced voltage, primary resistor
AC - Reduced voltage, primary reactor
WD - Reduced voltage, wye-delta
RR - Reduced voltage, resistor (dc)
PW - Part winding

1.2.1.4 Size. Size of the controller shall be designated by the numerical value (see 3.21.1, tables IV and V).

1.2.1.5 Low voltage protective feature. The protective feature shall be designated by the corresponding letter symbol as follows:

P - Low voltage protection
R - Low voltage release

1.2.2 Contactor. Contactors shall be designated in the following form and as specified (see 6.2 and 6.11).

1.2.2.1 Poles. The number of poles or main contacts shall be designated by the corresponding numerical value.

1.2.2.2 NO AUX contacts. The number of normally open auxiliary contacts shall be designated by the corresponding numerical value.

1.2.2.3 NC AUX contacts. The number of normally closed auxiliary contacts shall be designated by the corresponding numerical value.

1.2.3 Associated switching device and controller part. Associated switching devices and controller parts shall be designated in the following form and as specified (see 6.2 and 6.11).

1.2.3.1 Equipment type. The equipment type shall be designated by the corresponding letter symbols as follows:

ASD - Associated switching device
CTP - Controller part



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1.2.3.2 Device or part. The device or part shall be designated by the corresponding letter symbols as follows:

R - Relay
TDR - Time delay relay
CR - Current relay
CTR - Control relay
OLR - Overload
PS - Pressure switch
TS - Temperature switch
LS - Limit switch
SS - Selector switch
PBS - Pushbutton switch
XS - Transfer switch
PXS - Proximity switch
PST - Pushbutton station
MS - Master switch
FS - Float switch
DS - Drum switch

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

J-W-1177 - Wire, Magnet, Electrical, General Specification.

MILITARY

MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036 - Enclosures for Electric and Electronic Equipment.
MS3400 - Connectors, Receptacle, Electric, Wall Mounting, Front Release, Crimp Contact, AN Type.
MIL-L-3661 - Lampholders, Indicator Lights, Indicator-Light Housings, and Indicator-Light Lenses, General Specification for.
MIL-L-3661/63 - Lampholder, Lights, Indicator (Housing), Style LH96.
MIL-L-3661/65 - Lampholder, Lights, Indicator (Housing), Style LH98.



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MILITARY (Continued)

| | |
|---|---|
| MIL-R-15109 | - Resistors and Rheostats, Naval Shipboard. |
| MIL-F-15160 | - Fuses: Instrument, Power, and Telephone. |
| MIL-F-15160/60 | - Fuses; Instrument, Power and Telephone (Nonindicating), Style F60. |
| MIL-F-15160/61 | - Fuses; Instrument, Power and Telephone (Nonindicating), Style F61. |
| MIL-S-15291 | - Switches, Rotary, Snap Action and Detent/Spring Return Action, General Specification for. |
| MIL-T-16315 | - Transformers, Power, Step-Down (Miscellaneous, Naval Shipboard Use). |
| MIL-W-16878 | - Wire, Electrical, Insulated, General Specification for. |
| MIL-M-17059 | - Motors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use). |
| MIL-M-17060 | - Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use. |
| MIL-E-17555 | - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of. |
| MIL-S-18396 | - Switches, Meter and Control, Naval Shipboard. |
| MIL-S-19500 | - Semiconductor Devices, General Specification for. |
| MIL-R-19523 | - Relays, Control, Naval Shipboard. |
| MIL-S-19622 | - Stuffing Tubes, Nylon; and Packing Assemblies; General Specification for. |
| MIL-F-21346 | - Fuseholders, Block, and Shroud Type, and Associated Fuse Clips; General Specification for. |
| MIL-M-38510 | - Microcircuits, General Specification for. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-108 | - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment. |
| MIL-STD-167-1 | - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited). |
| MIL-STD-202 | - Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-461 | - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference. |
| MIL-STD-462 | - Electromagnetic Interference Characteristics, Measurement of. |
| MIL-STD-740-1 | - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment. |
| MIL-STD-740-2 | - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment. |
| MIL-STD-883 | - Test Methods and Procedures for Microelectronics. |
| MIL-STD-1399, Section 300 | - Interface Standard for Shipboard Systems Electric Power, Alternating Current. (Metric) |
| MIL-STD-1683 | - Connectors and Jacketed Cable, Electric, Selection Standard for Shipboard Use. |



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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

TE000-AB-GTP-0010 - Parts Application and Reliability Information Manual for Navy Electronic Equipment.

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standards for Industrial Controls and Systems. (DoD adopted)

ICS 2 - Standards for Industrial Control Devices, Controllers and Assemblies. (DoD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 "L" Street, NW, Washington, DC 20037.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Contactors, relays, switches, and controller enclosures furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.5).

3.2 Controllers. Controllers provided under this specification shall use contactors, relays, controller enclosures and switches (as applicable) which are qualified for listing on the applicable qualified products list at the time of award of contract (see 3.1).



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3.3 Basic requirements. Unless otherwise specified herein, the starter shall conform to the requirements of MIL-E-917 for the following items (see 6.3):

(a) Materials.
(b) Electrical shock hazard.
(c) Electrical creepage and clearance distances.
(d) Threaded parts and fastening devices.
(e) Electrical insulation.
(f) Insulating procedures.
(g) Wire, wiring methods and marking.
(h) Parts.
(i) Processes.

3.4 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.5 Power.

3.5.1 Dc controllers.

3.5.1.1 Surface ships. Dc controllers shall operate between 80 and 110 percent of rated voltage at the specified ambient temperature.

3.5.1.2 Submarines. Dc controllers shall operate at the specified ambient temperature between 180 and 355 volts (nominal 250 volts direct current (Vdc)) or between 350 to 710 Vdc (nominal 500 Vdc).

3.5.1.3 Dc controllers. Dc controllers having a motor speed adjustment by field rheostat exceeding a speed range of 1-1/2 to 1 shall provide full field starting and field acceleration features.

3.6 Connection diagram and table of overload heaters and reactors. Each controller shall include a connection diagram, a schematic diagram, a description of operation, and a table of overload heaters, coils, and reactors (if used). The information shall be printed by a process that is nonfading, protected by transparent plastic, and secured inside of the enclosure.

3.7 Duty. Unless otherwise specified, controllers shall be constructed for continuous duty and general purpose service. For special duty such as jogging or plugging, constant horsepower multispeed motors, lighting or transformer loads, special ratings may be required. Requirements for such service will be determined by the contracting activity and shall be as specified (see 6.2).



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3.8 Circuit arrangement. Controllers shall operate on an ungrounded system. Each controller shall be electrically independent, except for interlocks, and will normally derive its power from the same source as the controlled device. A defect in one controller shall not disable any other controller. Current transformer secondaries shall be grounded to the controller enclosure. Main line fuses, disconnects or circuit breakers shall not be provided as part of the controller. Contacts shall not be connected in parallel to obtain higher current carrying capacity. Unless otherwise specified (see 6.2), control circuits shall be the same voltage as the power circuit. Line contacts of one contactor shall not be used to maintain power to another contactor unless tests, in accordance with 4.8.5, have been made to ensure that a shock will not cause the maintained contactor to drop out. Multispeed and reversing controllers shall be arranged so that speed and direction changes are effected by a single operation. It shall not be possible for a motor to be in a different mode of operation than indicated by the associated switching device's identification plate. When timing relays are used to bypass safety switches during the starting period, the circuit shall be wired so that an opening in the relay circuit will prevent starting of the motor. The stop device of a controller shall be so connected that the controlled device will be de-energized and can be started again only upon actuating the start device.

3.9 Reduced voltage controllers. Reduced voltage controllers shall provide closed circuit transition.

3.10 Load isolation. If specified (see 6.2), opening of the line contactors shall isolate all portions of the load from the line except control circuits and the incoming power terminals. A discharge circuit shall be provided for shunt field winding dc controllers.

3.11 Remote device circuits. Control circuit leads from controllers to remote devices shall be protected against damage from short circuits. Leads wired in series with current limiting parts, such as coils and resistors, which are located within the enclosure will be considered to be adequately protected. Leads not protected by current limiting parts within the enclosure shall be fused. When specified (see 6.2), remote indicating and monitoring circuits shall be independently fused so that a short circuit in these circuits will not prevent the controller from performing its function. When fuses larger than 10-ampere capacity are required, 15-ampere fuses shall be used.

3.12 Ambient temperature ratings. The ambient temperature ratings for the motor controller shall be in accordance with 3.12.1 and 3.12.2, as specified (see 6.2).

3.12.1 Rating 50 degrees Celsius (°C). The rating of motor controllers shall be based on an ambient temperature of 50°C for those installations where the maximum normal operating temperature of the surrounding atmosphere or other cooling medium is 50°C or less. Motor controllers constructed for use in a 50°C ambient temperature shall withstand exposure to an ambient temperature of 70°C without damage for 1 hour. During the exposure at 70°C, the motor controllers may be energized but they need not operate as specified herein. After the 1 hour period when they have cooled to a temperature within their operating range, they shall operate as specified herein.



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3.12.2 Rating 65°C. The rating of motor controllers shall be based on an ambient temperature of 65°C for those installations where the maximum normal operating temperature of the surrounding atmosphere or other cooling medium is 65°C or less but more than 50°C. Motor controllers constructed for use in a 65°C ambient temperature shall withstand exposure to an ambient temperature of 85°C without damage for 1 hour. During the exposure at 85°C, the motor controllers may be energized but they need not operate as specified herein. After the 1 hour period when they have cooled to a temperature within their operating range, they shall operate as specified herein.

3.13 Wiring. Permanent internal wiring shall be copper wire or bus bar. The size, stranding, and insulation of wire, and the dimensions, insulation, and spacing of bus bars shall be mechanically and electrically suited to the application. Magnet wire shall conform to J-W-1177. Wires connecting to resistors or class H coils, and wires otherwise exposed to heat shall be a crosslinked modified polyethylene (XLPE) insulation wire with a rating of 125°C, minimum, as specified in MIL-W-16878. The minimum wire size shall be AWG 14, except that AWG 16 may be used for connections to local switches and that AWG 22 may be used for all connections to the programmable logic controllers.

3.13.1 External cable connections. For the external conductors entering a controller, connections shall be made to terminal boards or studs located within the enclosure near the cable entrance. Terminal boards and studs shall be accessible from the front of the enclosure. Circuitous routing of conductors shall be avoided.

3.14 Mounting. Parts mounted on panels shall be replaceable without removing the panel, except for low noise controllers (see 3.26.16 and 6.2).

3.15 Endurance. Endurance requirements for manual controllers, contactors, switches, and relays (except dc contactors, overload relays, and relays conforming to MIL-R-19523) shall consist of two series of electrical operations plus one series of mechanical operations, as shown in table I. The overload relay endurance requirements shall be the series of electrical operations shown in table I.



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TABLE I. Endurance requirements.

| Item | No. of operations 1/ | Percent of rated current | | At percent rated voltage | Power-factor (lagging) |
|---|---|---|---|---|---|
| | | Make | Break | | |
| Manual controllers (ac) | 25,000 | 600 | 100 | 100 | 1 |
| Manual controllers (ac) | 50 | 600 | 600 | 110 | 0.4 to 0.5 |
| Manual controllers (ac) | 25,000 | Mechanical operations | | | |
| Manual controllers (dc) | 25,000 | 400 | 100 | 100 | |
| Manual controllers (dc) | 50 | 400 | 400 | 110 | |
| Manual controllers (dc) | 25,000 | Mechanical operations | | | |
| Contactors (ac): | | | | | |
| Size 0 through 5 | 50,000 | 600 | 100 | 100 | 1 |
| Size 6 | 25,000 | 600 | 100 | 100 | 1 |
| Size 0 through 6 | 50 | 600 | 600 | 110 | 0.4 to 0.5 |
| Size 0 through 4 | 1,000,000 | Mechanical operations | | | |
| Size 5 and 6 | 300,000 | Mechanical operations | | | |
| Contactors (dc): | | | | | |
| Size 0 through 6 | 25,000 | 400 | 100 | 100 | |
| Size 0 through 6 | 50 | 400 | 400 | 110 | |
| Relays (except overload and relays conforming to MIL-R-19523) | Ac - same requirements as for size 0 ac contactors<br>Dc - same requirements as for size 0 dc contactors | | | | |
| Switches | 50,000 | 100 | 100 | 100 | |
| | 50 | 150 | 150 | 100 | |
| | 50,000 | Mechanical operations | | | |
| Overload relays | 500 | 100 | 100 | 100 | 1 |

1/ One operation means one electrical make and break for each set of contacts.

3.16 Detail requirements.

3.16.1 Manual controllers. Size 1 manual controllers shall be arranged so that the overload device is defeated by holding down the start button. Emergency-run features shall be not be provided on size 0 manual controllers.



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3.16.2 Interlocks. Mechanical and electrical interlocks shall be provided between reversing or speed selection contactors to eliminate the possibility of a short circuit resulting from simultaneous closing of contactors.

3.16.3 Emergency run. When specified (see 6.2), magnetic controllers shall be provided with a means of defeating the overload relay for emergency running. This feature may be combined with the overload relay reset, or obtained with a separate push-button. Manual operation of the emergency run switch shall be required for the duration of the emergency. Emergency run shall not be combined with the start push-button.

3.16.4 Local transfer. When specified (see 6.2), controllers that are intended for use with remote switches shall have a local transfer switch. This local transfer switch shall have a start push-button shunted across the contactor coil which will disconnect the remote control wiring and allow local control. This switch shall prevent a damaged remote switch or defective wiring from disabling the controller. Generally, a local transfer switch is required only for vital functions such as steering and fire pump control.

3.16.5 Contactors and relays. Coils shall be encapsulated as specified in MIL-E-917. Coil leads shall terminate at screw terminals. Contacts shall be of simple and rugged construction and shall be easily replaceable. The construction shall be such that slight projections which may occur during service shall not prevent proper operation of the contacts. Springs shall be constructed to ensure the proper functioning of the relays and contactors. Springs shall not be stressed beyond their fatigue limits. The design shall be such that springs are not depended upon to carry current. Contacts, other than sliding contacts in control circuits, shall not be plated, except with silver. Contactors shall incorporate a captive fastener at each mechanical joint to prevent the hex head terminal bolt from rotating. Coils of the contactors and control relays shall be shunted by a varistor and an R-C network combination that shall prevent contact arcing. The contact assembly, as well as other current carrying parts of time delay relays, may be plated with electrodeposited gold for corrosion resistance purposes. Mating armature faces shall not be plated, painted, or coated in any manner that will interfere with correct electrical operation. Contactors and relays shall meet the endurance requirements specified in table I.

3.16.5.1 Relays. Relays shall conform to MIL-R-19523 and the applicable requirements of this specification.

3.16.6 Overload protection. General purpose overload relays shall be used to protect continuous duty, intermittent duty, and varying duty constant speed or multi-speed ac motors constructed in accordance with MIL-M-17059 or MIL-M-17060. Overload relays shall have tripping characteristics falling within the acceptable band shown on figure 1. When tripped at 150 percent of rated current in an ambient temperature of 20 to 30°C, overload relays shall be resettable within 2 minutes after tripping. Overload relays shall be constructed to trip within 3 hours when carrying rated relay current. The overload relay rating shall be 115 to 125 percent of the motor identification plate current. The overload relay rating for magnetic controllers shall change not more than 3 percent for each 10°C change in ambient temperature in the range between 20 and 70°C. For manual



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controllers, the rating of the overload relay shall change not more than 5 percent for each 10°C change in ambient temperature in the range between 20 and 70°C. When specified (see 6.2), direct temperature sensing overload protection systems shall be used to protect variable speed motors (either variable voltage or variable voltage and frequency control systems) and motors where the duty cycle may be worsened by operator control or by operational mode which exceeds motor thermal limits because of mechanical or electrical characteristics of the motor or its load. Overload protection systems using temperature sensing devices in the motor winding shall be compatible with the temperature sensors. This system shall energize an alarm or de-energize the motor control. The system shall be fail safe. Overload relays shall be calibrated for operation in the specified ambient temperature. Overload relays shall meet the endurance requirements specified in table I.

3.16.7 Switches. Switches shall conform to the applicable specifications MIL-S-15291, MIL-S-18396, or applicable requirements of this specification. Endurance requirements of switches shall be as specified in table I. Each position of manual switches, other than spring return switches, shall have definite positioning that can be felt by the operator. Switch identification plates shall clearly indicate each position. Switches shall be mountable in any position. Sensing bulbs of temperature control shall be mounted in the position specified by the manufacturer. When specified (see 6.2), pressure switches shall be supplied with ambient pressure compensation. Pressure switches shall withstand a pressure of at least three times the maximum operating pressure, but no longer need to be operational.

3.16.8 Transformers. Control circuit transformers shall be rated and shall conform to MIL-T-16315. Devices powered by the transformer shall operate satisfactorily from Navy type I power and shall meet the operational requirements of this specification. Autotransformers shall be used only for reduced-voltage rating.

3.16.9 Programmable logic controllers (PLC). When specified (see 6.2), a PLC shall be installed within the motor controller enclosure and shall perform normal programmed functions without operator interface.

3.16.9.1 Memory. All PLCs shall use either an ultra-violet or electrically erasable Programmable Read Only Memory (PROM) as a non-volatile program storage medium. The PROM shall be replaceable with a minimum of controller disassembly and shall not be programmable while installed in the PLC. Upon application of power to the PLC, programming contained in the PROM shall be downloaded into a Random Access Memory (RAM) for program execution. The RAM shall be a maximum of 2048 bytes. Replacement of the PROM shall initiate reloading of the operating program into RAM. The RAM shall have power source protection to prevent memory alteration during short-term (30 minutes or less) interruption of the normal power supply. Such protection shall be either a rechargeable battery or capacitor. The PLC shall be unaffected by automatic bus transfers which cause momentary power loss to the motor controller.

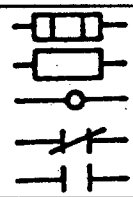
3.16.9.2 Programming. The PROM shall be programmable using an IBM PC or compatible microcomputer operating in an MS-DOS environment using manufacturer-supplied software and a cable interface capable of being attached to a standard serial RS-232 communications port. It shall not be necessary to use any expansion



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cards or other hardware mounted internal to the microcomputer to perform programming functions, and it shall not be possible to alter the PROM program while the PLC is mounted in the motor controller enclosure. Programming shall be performed using relay ladder logic and the symbols defined in table II.

TABLE II. Relay ladder logic programming symbols.

| Item | Symbol |
|---|---|
| Contact (NO)<br>Contact (NC)<br>Coil<br>Timer<br>Counter |  |

3.16.9.3 Operational testing. Portable or built-in test equipment shall be provided that is sufficient to allow the operator to isolate any PLC malfunction to the lowest replaceable element of the PLC with a minimum of disassembly and to perform any required periodic testing. Periodic calibration shall not be a requirement for PLC continued performance. To the maximum extent, testing shall be automatically performed without interference with normal PLC operation.

3.16.9.4 Interchangeability. Components of the PLC shall be fully interchangeable with other PLCs of a given manufacturer. Components shall not be designed to serve a single unique application unless specifically authorized by the Government. The installation envelope allowed for a PLC within the enclosure shall be as shown on figure 2. The enclosure shall allow the space shown on figure 2 for PLC installation whether or not the PLC actually used will require the entire space. Unless otherwise specified (see 6.2), the installation footprint for PLCs in the enclosure shall be as shown on figure 3. All PLCs in each size controller shall use an electrical connector, MS3400 type, for interfacing all input and output signals with ship's wiring. Wiring shall be in accordance with MIL-STD-1683. Pins for these connectors shall be dedicated as indicated in the purchase specification. Cable for connecting the PLC to the above connector shall be supplied by the manufacturer. All wiring within the enclosure (for example, power supply, contactor interface) shall be field run. Special signal requirements (for example, thermocouples) may be excepted from the connector requirement.

3.16.9.5 Parts selection and screening. Unless otherwise specified (see 6.2), parts selection and application shall be in accordance with NAVSEA TE000-AB-GTP-0010. In addition, the following shall apply:

(a) The maximum junction temperatures for solid state components shall not exceed 105°C in the worst case conditions of component or ambient temperatures. A minimum of 50°C part ambient temperature shall be assumed.



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(b) Established Reliability (ER) active/passive electrical components shall be used in the manufacture of the PLC where possible. Level R shall be used for passive components, JANTX shall be used for semiconductors, and MIL-M-38510, class B, shall be used for microcircuits. Where semiconductors or microcircuits are approved which cannot be acquired in accordance with MIL-S-19500 or MIL-M-38510, then semiconductors shall receive screening equivalent to the JANTX counterpart, and microcircuits shall be screened to the conditions defined by MIL-STD-883, class B.

(c) Subassemblies of the PLC shall be Environmentally Stress Screened (ESS) as an integral part of the manufacturing process.

3.16.9.6 Repairability. The PLC shall be of a modular design that allows troubleshooting and module replacement without removing the PLC from the motor controller enclosure and with a minimum of internal disassembly. Isolation and replacement of PLC hardware failures shall require not more than 15 minutes.

3.16.9.7 Power. The PLC shall not require a dedicated power source external to the motor controller, and all power conditioning required for PLC operation (for example, transformers and filters) shall be contained within the envelope defined in 3.16.9.4. The PLC shall operate from power already available within the enclosure, and shall be self-compatible and compatible with peripheral equipments. Switching functions and internally generated electromagnetic interference (EMI) shall not be propagated outside the motor controller enclosure by way of input or output interface signals or cause inadvertent alarms or other out-of-tolerance perturbations. The PLC shall meet the EMI requirements of MIL-STD-461. The PLC shall not be susceptible to damage due to transients and excursions of voltage, current, or frequency common to motor controller operation. Power up, power down, and power interrupt characteristics shall be predictable, repeatable, and fully defined by application information prepared by the manufacturer. For those inputs which require excitation power, such as a 4-20 milliampere (mA) transducer, the power supply shall be an integral part of the PLC.

3.16.9.8 Input and output. The PLC shall process the types of input and output signals specified in table III and as specified (see 6.2). This may be accomplished by using interchangeable modules or expansion units so long as these fit within the envelope defined in 3.16.9.4.

TABLE III. Input and output types for PLCs.

| Signal type | Analog (A) Discrete (D) | ac/dc | Voltage/amperage |
|---|---|---|---|
| Input | D | dc | 0/5 Vdc |
| | D | dc | 0/10 Vdc |
| | D | dc | 0/28 Vdc |
| | D | dc | 0/115 Vdc |
| | D | dc | 0/230 Vdc |
| | D | ac | 0/115 Vac |
| | A | dc | 0-5 Vdc |
| | A | dc | 0-10 Vdc |
| | A | dc | -5 to +5 Vdc |
| | A | dc | 4-20 mA |



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TABLE III. Input and output types for PLCs - Continued.

| Signal type | Analog (A) Discrete (D) | ac/dc | Voltage/amperage |
|---|---|---|---|
| Output | D | dc | 0/5 Vdc |
| | D | dc | 0/28 Vdc |
| | D | dc | 0/115 Vdc |
| | D | ac | 0/115 Vac |
| | A | dc | 0-10 Vdc |
| | A | dc | -5 to +5 Vdc |

3.17 Space heaters. When specified (see 6.2), controllers and switches in watertight or submersible enclosures with an enclosed volume larger than 150 cubic inches ($in^3$) shall be provided with space heaters to prevent condensation. Space heaters shall be dual voltage for use on 115 or 230 volts alternating current (Vac) or Vdc.

3.18 Indicator lights. If indicator lights are required (see 6.2), ac indicator lights shall conform to MIL-L-3661 and MIL-L-3661/65 for 450 volt service and 120 volt service. Dc indicator lights shall conform to MIL-L-3661/63.

3.19 Fuses and fuseholders. Fuses shall conform to style F60 of MIL-F-15160 and MIL-F-15160/60. Existing controller designs may continue use of style F61 fuses of MIL-F-15160/61. Fuseholders and associated clips shall be in accordance with MIL-F-21346, except all fuse clips shall be the steel copper-clad type (material designation F). The fuse type designation specified in MIL-F-15160 shall be permanently marked adjacent to each fuse block or fuseholder when space permits; otherwise, it shall be permanently marked in a conspicuous location.

3.20 Resistors and rheostats. Resistors and rheostats shall be in accordance with MIL-R-15109.

3.21 Characteristics. The following characteristics of controllers shall be as specified (see 6.2):

(a) Voltage rating (see 6.2):

115 volts, single-phase
115 volts, 3-phase
440 volts, 3-phase
115 Vdc
230 Vdc
250 Vdc (nominal) range 180 to 355 V, submarine service
500 Vdc (nominal) range 350 to 710 V, submarine service (special) (see 6.2)

(b) Current rating: Current rating shall be as required by the applicable motor size specified in 3.21.1 (see 6.2).



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(c) Manual controllers (see 6.2):

Type: Across-the-line
Protection: Overload relay
Function: Motor starting (low voltage release effect)

(d) Magnetic controllers (see 6.2):

Type: Across-the-line
Reduced voltage
(1) Autotransformer
(2) Primary resistor
(3) Primary reactor
(4) Wye-delta
(5) Resistor (dc)

Part winding

Protection: Low voltage protection (LVP)
Low voltage release (LVR)
Overload relay
Temperature sensor actuated

Function: Motor starting
Motor starting and reversing
Motor starting and speed regulation
Motor starting and speed selection
Motor starting, reversing, and speed selection
Miscellaneous

(e) Switches (see 6.2):

| Type: | Manual | Automatic |
|---|---|---|
| | Pushbutton | Pressure |
| | Rotary selector | Thermostatic |
| | Drum | Limit |
| | | Float |
| | | Proximity |

Function: Usually to actuate a magnetic controller
Location: Local
Remote

3.21.1 Horsepower (hp) ratings. Controllers shall be provided in the sizes, and shall conform to the applicable hp and current rating, shown in tables IV and V.



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TABLE IV. Hp rating for dc contactors and controllers (see 6.2).

| Hp ratings of reduced-voltage magnetic controllers | | | |
|---|---|---|---|
| Size of controller | Continuous rating amperes | 115 V hp rating | 230 V hp rating |
| 1 | 25 | 3 | 5 |
| 2 | 50 | 5 | 10 |
| 3 | 100 | 10 | 25 |
| 4 | 150 | 20 | 40 |
| 5 | 300 | 40 | 75 |
| 6 | 600 | 75 | 150 |
| 7 | 900 | 110 | 225 |

| Hp rating of full-voltage magnetic controllers | | | |
|---|---|---|---|
| Size of controller | Continuous rating amperes | hp at | |
| | | 115 V | 230 V |
| 0 | 15 | 1 | 1 |
| 1 | 25 | 1-1/2 | 2 |

TABLE V. Hp rating for ac contactors and controllers (see 6.2).

| Rating of polyphase single-speed full-voltage magnetic controllers for nonplugging and nonjogging duty | | | |
|---|---|---|---|
| Size of controller | Continuous current rating amperes | Three-phase hp at | |
| | | 115 V | 440 V |
| 0 | 18 | 2 | 5 |
| 1 | 27 | 3 | 10 |
| 2 | 45 | 7-1/2 | 25 |
| 3 | 90 | 15 | 50 |
| 4 | 135 | 30 | 100 |
| 5 | 270 | - | 200 |
| 6 | 540 | - | 400 |
| 7 | 810 | - | 600 |



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TABLE V. Hp rating for ac contactors and controllers - Continued.

| Ratings for single-phase full-voltage magnetic controllers for nonplugging and nonjogging duty | | |
|---|---|---|
| Size of controller | Continuous current rating amperes | Single-phase hp at 115 V |
| 0 | 18 | 1 |
| 1 | 27 | 2 |
| 2 | 45 | 3 |

| Ratings of polyphase reduced-voltage, general purpose, reversing or nonreversing magnetic controllers | | | |
|---|---|---|---|
| Size of controller | Continuous current rating amperes | Three-phase hp at | |
| | | 115 V | 440 V |
| 1 | 27 | 3 | 10 |
| 2 | 45 | 7-1/2 | 25 |
| 3 | 90 | 15 | 50 |
| 4 | 135 | 30 | 100 |
| 5 | 270 | -- | 200 |
| 6 | 540 | -- | 400 |
| 7 | 810 | -- | 600 |

| Ratings for single-phase, reduced-voltage, general purpose, reversing or nonreversing magnetic controllers | | |
|---|---|---|
| Size of controller | Continuous current rating amperes | Single-phase hp at 115 V |
| 1 | 27 | 2 |
| 2 | 45 | 3 |

| Ratings of manual controllers for single-speed and multispeed motors | | | | |
|---|---|---|---|---|
| Size of controller | Three-phase | Single-phase | | Continuous current rating amperes |
| | 115 V | 440 V | 115 V | |
| 0 | 1-1/2 | 2 | 1 | 15 |
| 1 | 3 | 7-1/2 | 1-1/2 | 25 |



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TABLE V. Hp rating for ac contactors and controllers - Continued.

| Ratings for wye-delta magnetic controllers for nonreversing, nonjogging duty for open or closed-circuit transition | | |
|---|---|---|
| Size of controller | Continuous current rating amperes | Three-phase hp at 440 V |
| 1 | 47 | 15 |
| 2 | 78 | 40 |
| 3 | 156 | 75 |
| 4 | 233 | 150 |
| 5 | 467 | 300 |
| 6 | 935 | 700 |
| 7 | 1400 | 1000 |

| Ratings for nonreversing, nonjogging, reduced-voltage, polyphase magnetic controllers for motors for part-winding starting | | |
|---|---|---|
| Size of controller | Continuous current rating amperes | Three-phase hp at 440 V |
| 1 | 54 | 20 |
| 2 | 90 | 50 |
| 3 | 180 | 100 |
| 4 | 270 | 200 |
| 5 | 540 | 400 |

| Ratings for multispeed, magnetic controllers for nonplugging and nonjogging duty for constant-horsepower motors | | | |
|---|---|---|---|
| Size of controller | Continuous current rating amperes | Three-phase hp at | |
| | | 115 V | 440 V |
| 0 | 18 | 1 | 3 |
| 1 | 27 | 2 | 7-1/2 |
| 2 | 45 | --- | 20 |
| 3 | 90 | --- | 40 |
| 4 | 135 | --- | 75 |
| 5 | 270 | --- | 150 |
| 6 | 540 | --- | 300 |
| 7 | 810 | --- | 450 |



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3.22 Enclosures. Enclosures shall be in accordance with MIL-E-2036 and MIL-STD-108 and shall be one of the following types, as specified (see 6.2):

(a) Dripproof (45 degrees)
(b) Watertight
(c) Submersible (15 feet)
(d) Explosion-proof (class 1, group D)
(e) Splashproof

3.22.1 Special provisions for enclosures. If the degree of enclosure is not specified, dripproof (45 degrees) enclosures shall be used for controllers, and watertight enclosures shall be used for associated switching devices. Watertight and submersible enclosures larger than 60 inches shall have a moisture drain in the bottom. The opening shall be closed with a 1/4-inch pipe plug. Enclosures shall be constructed so that all necessary cables can be brought into either the top or bottom. Enclosures larger than 4 cubic feet shall have removable cable entrance plates. Each enclosure shall be constructed to support itself and its parts when deck or bulkhead mounted. Fans shall not be used. Enclosures shall be constructed to allow cable entrance by one of the following methods:

(a) Stuffing tube in accordance with MIL-S-19622.
(b) Female National Taper Pipe Thread (NPT).

3.22.2 Large enclosures. Enclosures larger than 1.5 cubic feet, except explosion-proof enclosures, shall have hinged doors. Doors wider than 18 inches shall be removable, and the removal shall require not more than 18 inches of free space in front of the enclosure. Doors on dripproof enclosures shall be secured by captive thumb screws or other fasteners with all parts captive. Doors on watertight and submersible enclosures shall be secured by captive screws.

3.22.3 Small enclosures. Enclosures 1.5 cubic feet and smaller may have hingeless removable covers provided that no electrical devices are mounted on the cover.

3.22.4 Plastic enclosures. Plastic enclosures shall have any penetration of the enclosure completely insulated from possible electrical contact with internal conductors.

3.22.5 Aluminum enclosures. Aluminum enclosures shall have any penetration of the enclosure completly insulated from possible electrical contact with internal conductors.

3.22.6 Dimensional limits. Dimensional limits of the enclosures shall be in accordance with table VI.



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TABLE VI. Dimensions of enclosures.

| Across-the-line | | | |
|---|---|---|---|
| Size | Height (inches) | Width (inches) | Depth (inches) |
| 0 | 25 | 20 | 10 |
| 1 | 25 | 20 | 10 |
| 2 | 25 | 20 | 10 |
| 3 | 30 | 25 | 10 |
| 4 | 30 | 25 | 10 |
| 5 | 35 | 30 | 15 |
| 6 | 35 | 30 | 15 |
| 7 | 35 | 30 | 15 |
| Reduced-voltage | | | |
| Size | Height (inches) | Width (inches) | Depth (inches) |
| 0 | 45 | 30 | 15 |
| 1 | 45 | 30 | 15 |
| 2 | 45 | 30 | 15 |
| 3 | 45 | 30 | 15 |
| 4 | 65 | 30 | 30 |
| 5 | 95 | 35 | 30 |
| 6 | 95 | 35 | 30 |
| 7 | 95 | 35 | 30 |

3.23 Safety. Electrical safety shall be in accordance with the requirements of MIL-E-917.

3.24 Painting. Parts not constructed of corrosion-resisting material, or having a corrosion-resistant treatment as specified in MIL-E-917, shall be painted as specified therein. Touching-up is permitted for marks or scratches due to assembly, testing, or other handling.

3.25 Identification plates. Each controller shall have an identification plate containing the manufacturer's name, catalog number, national stock number (NSN), if specified (see 6.2), voltage, frequency, and maximum hp rating, and low noise, if applicable. Each associated switching device shall have identification containing sufficient information to identify and reorder the item with information available in the controller technical manual. Preferred location of the information is inside the enclosure of the controller and on the outside of switch housings. Identification plates shall be not less than 0.020 inch in thickness, and shall be brass, corrosion-resistant steel, or anodized aluminum marked by etching, engraving, stamping, or photo-processing. Etching, engraving, or stamping shall be not less than 0.003 inch deep. Identification plates, when used for associated devices, shall be securely fastened to the device. Where metal fasteners are not used, adhesive backed metal (not less than 0.010 inch in thickness) of plastic film, type G as specified in MIL-I-631 may be used. Items that cannot be provided with a plate, such as tubular ceramic resistors, may be



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marked by stamping with a permanent ink and covered by a protective coating that shall withstand the temperature characteristics the part will exhibit. Controllers and switches shall be permanently and clearly marked with an identification plate showing the function of each position subject to manual operation. Controllers having more than one source of power shall be provided with an identification plate warning that all power sources shall be disconnected before working on the controller. The word "WARNING" shall be in prominent red letters. The plate shall be mounted on the front of the enclosure. When the controlled device is protected by special circuitry in the controller, the identification of the associated sensor shall be clearly marked on an identification plate on the controller. Laser marking of nameplates may be applied.

3.26 Performance.

3.26.1 Ac controllers. Ac controllers shall operate on type I, 60-hertz (Hz) power as specified in MIL-STD-1399, section 300 and shall operate between 80 and 110 percent of rated voltage at the specified ambient temperature.

3.26.1.1 Ac input. The ac input to the controller shall be 440 V, 60-Hz, 3-phase, ungrounded, type I power in accordance with the steady-state and transient characteristics as specified in MIL-STD-1399, section 300.

3.26.2 Overload protection. Overload protection shall be provided for the motor during starting and running conditions. Resetting, if not automatic, shall be accomplished by means external to the enclosure. When two single-pole relays are used with 3-phase motors, resetting of the two relays shall be accomplished by a single button. Automatic reset overload relays shall not be used with controllers arranged for LVR. When an overload protection system requires external power, the power source shall be the motor power line or the motor control circuit. Overload protection shall not be provided in dc controllers for submarine service; however, provision shall be made for wiring to an overspeed trip module, if required (see 6.2).

3.26.3 Temperature rise. Temperature rises shall not exceed the values shown in table VII. Coil rises shall be determined by the resistance method. All other rises shall be measured by thermocouple or thermometer. Coils energized by the emergency-run feature shall be energized for 30 minutes without exceeding a total temperature of 200°C.

TABLE VII. Maximum temperature rise.

| Part | Allowable temperature rise over ambient (°C) | |
|---|---|---|
| | 50°C ambient | 65°C ambient |
| Shunt coils: | | |
| Class A insulation | 60 | 45 |
| Class B insulation | 85 | 70 |
| Class F insulation | 110 | 95 |
| Class H insulation | 135 | 120 |
| Series coils: Single layer, Class A insulation | 50 | 35 |



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3.26.4 Shock. Equipment shall conform to shock requirements for grade A, class I, type A, light or medium weight equipment (as applicable) specified in MIL-S-901, and the following:

(a) Line contacts. Unless otherwise specified (see 6.2), closed line contacts shall not open for more than 10 milliseconds. Open line contacts shall not close for more than 10 milliseconds.
(b) Auxiliary contacts and contacts on associated switching devices. Open contacts shall not close for more than 10 milliseconds. Closed contacts shall not open for more than 10 milliseconds.
(c) Contactors and relays. Contactors and relays shall not drop out during shock when energized and held in through any of their normally open auxiliary contacts, and, if in the de-energized state, shall not pick up through the same contact during shock.
(d) Overload relays. Overload relays shall not trip when carrying up to 85 percent of the ultimate tripping current.
(e) Reduced voltage starters. Unless otherwise specified (see 6.2), timing relays in reduced voltage starters may recycle provided that the starter returns to normal running condition without operator intervention.

3.26.4.1 Resilient mounting. Resilient mounting of equipment shall be in accordance with MIL-S-901, grade A, class II, type A, light or medium weight when momentary opening of closed contacts, or momentary closing of open contacts, as specified in 3.26.4 (a) and (b) is not acceptable (see 6.2).

3.26.5 Vibration. Equipment vibration frequency shall be from 4 to 33 Hz and shall withstand type I tests of MIL-STD-167-1 without mechanical damage, contact chatter, or other electrical malfunctioning. When specified (see 6.2), equipment which will be exposed to frequencies above 33 Hz shall be vibrated up through 50 Hz.

3.26.6 Insulation resistance. Insulation resistance shall be not less than 10 megohms. Insulation shall withstand the requirements of method 302 of MIL-STD-202.

3.26.7 Dielectric. Controllers shall withstand the dielectric test in accordance with NEMA ICS 1-109.

3.26.8 Terminal strength. The terminals of the contactors and control relays shall withstand the terminal strength requirements of MIL-STD-202.

3.26.9 Overall efficiency. The overall efficiency of the reduced-voltage starter shall be not less than 98 percent.

3.26.10 Effect on input waveform and harmonics. When the reduced-voltage starter is powered, and is supplying any load from no load to rated load at the specified voltage and frequency, the waveform distortion and harmonics feeding back to the input voltage shall be as follows:

(a) The total deviation factor shall be not more than 5 percent.
(b) Additional harmonic content of any single harmonic shall be not more than that of the original harmonic by 2 percent.



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3.26.11 Inclination. Controllers shall meet the following requirements for inclined operation:

(a) Surface ships. Equipment shall operate when permanently inclined in any direction at any angle up to 15 degrees from the normal operating position. Equipment shall be constructed to function without damage during temporary inclinations such as produced by ship's rolling of 45 degrees in any direction from the normal operating position.

(b) Submarines. Equipment shall operate when permanently inclined in any direction at any angle up to 30 degrees from the normal operating position. Equipment shall be constructed to function without damage during temporary inclinations such as produced by ship's rolling of 60 degrees in any direction from the normal operating position.

3.26.12 Environmental. The PLC shall meet the environmental requirements of MIL-STD-202.

3.26.13 EMI. The PLC shall meet the EMI requirements of MIL-STD-461.

3.26.14 Pick-up and drop-out. Ac contactors and relays shall pull in below 80 percent and drop out above 10 percent of rated voltage. Dc contactors and relays for surface ships shall pull in below 68 percent and drop out above 5 percent of rated voltage. Dc contactor and relays for submarines shall pull in below 160 Vdc and drop out above 13 Vdc.

3.26.15 Terminals and terminal boards. Power current carrying studs, screws, and other current carrying components of power connections shall be constructed of nonferrous materials. Control current carrying components or connections shall be constructed of either nonferrous materials or silver-plated ferrous materials. Construction materials shall be durable and corrosion-resistant.

3.26.16 Noise. When specified (see 6.2), ac controllers for low noise applications shall meet the structureborne noise requirements for type III equipment of MIL-STD-740-2 and airborne noise requirements for grade A3 and A12 equipment of MIL-STD-740-1.

3.27 Workmanship. The starter, including all parts and accessories, shall be free from burrs and sharp edges. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and sub-assemblies, wiring, welding and brazing, plating, riveting, finishes, machine operations and screw assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The



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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in MIL-E-917.

4.4 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. Qualification inspection shall consist of the examination and tests specified in table VIII, and shall be conducted in the order specified in 4.8. Two copies of the test report should be forwarded to NAVSEA. Failure to submit test reports within 30 days after the test period may result in the disapproval of the qualification inspection.



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TABLE VIII. Inspections.

| Test | Qualification | Quality conformance | Requirement | Test method |
|---|---|---|---|---|
| Controller: | | | | |
| Examination | X | X | 3.3 | 4.6 |
| Temperature rise | X | X | 3.26.3 | 4.8.1 |
| Vibration | X | X | 3.26.5 | 4.8.4 |
| Shock 1/ | X | X | 3.26.4 | 4.8.5 |
| Inclination | X | X | 3.26.11 | 4.8.7 |
| Effectiveness of enclosure | X | X | 3.22 | 4.8.10 |
| Input harmonic and waveform distortion (reduced-voltage) | X | | 3.26.10 | 4.8.14 |
| Efficiency (reduced-voltage) | X | | 3.26.9 | 4.8.15 |
| Voltage variation | X | X | 3.26.1 | 4.8.13 |
| Endurance | X | X | 3.15 | 4.8.2 |
| Contactor, relay, and switch: | | | | |
| Examination | X | X | 3.3 | 4.6 |
| Temperature rise | X | X | 3.26.3 | 4.8.1 |
| Vibration | X | X | 3.26.5 | 4.8.4 |
| Shock 1/ | X | X | 3.26.4 | 4.8.5 |
| Overload protection (overload relay) | X | X | 3.26.2 | 4.8.3 |
| General operation | X | X | 3.16 | 4.8.6 |
| Insulation resistance | X | X | 3.26.6 | 4.8.8 |
| Dielectric | X | X | 3.26.7 | 4.8.9 |
| Noise (ac contactor and relay) 2/ | X | | 3.26.16 | 4.8.11 |
| Pick-up and drop-out voltage (contactor and relay) | X | X | 3.26.14 | 4.8.12 |
| Terminal strength (contactor and relay) | X | X | 3.26.8 | 4.8.16 |
| Endurance | X | X | 3.15 | 4.8.2 |
| Programmable logic controller: | | | | |
| Examination | X | X | 3.3 | 4.6 |
| Temperature rise | X | X | 3.26.3 | 4.8.1 |
| Vibration | X | X | 3.26.5 | 4.8.4 |
| General operation | X | X | 3.16 | 4.8.6 |
| Insulation resistance | X | X | 3.26.6 | 4.8.8 |
| Dielectric | X | X | 3.26.7 | 4.8.9 |
| Environmental screening | X | X | 3.26.12 | 4.8.17 |
| EMI | X | X | 3.26.13 | 4.8.18 |

1/ Qualification and quality conformance shock tests of table VIII apply to controller parts and associated switching devices. Completely assembled controllers shall also undergo shock testing in accordance with 4.8.5.2.



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2/ When required (see 6.2).

4.4.1 Samples for qualification inspection. One to five samples of each contactor, relay, switch or controller for which qualification is desired shall be submitted for qualification testing.

4.4.1.1 Authorization. Prior to authorization of qualification tests, the contractor shall submit complete test procedures, mechanical and electrical drawings to NAVSEA for review and acceptance.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table VIII (see 6.3).

4.6 Examination. Equipment shall be examined to determine that the material, safety to operating personnel, and construction are in accordance with the requirements of this specification.

4.7 Order of tests. Unless otherwise specified (see 6.2), the tests for general operation, voltage pick-up and drop-out, voltage variation, terminal strength, insulation resistance, dielectric, and effectiveness of enclosure shall precede and follow shock and vibration tests. The endurance test shall precede the shock test.

4.8 Tests.

4.8.1 Temperature rise. Temperature rise tests shall be performed for quality conformance inspection on the first production run only. Temperature rise tests are not required on subsequent production runs unless a change is made in design. The test methods used and the precautions observed in conducting temperature rise tests shall be in accordance with NEMA ICS 1-109 and the following:

(a) Temperature rise tests shall be made under conditions equivalent to normal operating conditions at rated voltage, frequency, and load, with the duty specified to determine that the rises specified in 3.26.3 are not exceeded.
(b) Tests shall be made with equipment in its enclosure if enclosure is provided as part of the equipment.
(c) The ambient temperature in the testing area shall vary not more than 10°C during tests.

4.8.1.1 Reduced voltage controllers. Starting autotransformers, reactors, and resistors shall be tested to determine conformance to temperature rise limits specified in NEMA ICS 2-213 and 2-214.

4.8.2 Endurance. Endurance tests at the specified ambient temperature shall be conducted between 10 and 20 operating cycles per minute, except tests making and breaking 600 percent of rated current shall be conducted at 6 operating cycles per minute (closed 1 second, open 9 seconds). Input and output current, voltage, frequency, and power shall be recorded every 60 minutes at rated load. A malfunction causing shutdown or out of specified limit performance during this period shall constitute a failure requiring corrective action. A failure has not occurred if malfunction can be attributed to external causes such as ambient temperature over the specified temperature, variation of input power parameters



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beyond specified limits, operator's error, and other unusual conditions. Equipment shall meet the requirements specified in 3.15.

4.8.2.1 Restarting endurance test. Upon completion of corrective action after malfunction, the equipment shall be restarted and the test run according to the following conditions until a satisfactory operation endurance run is completed:

(a) If failure occurs during the first 25 percent of the endurance run, the test shall be restarted from the first operation, and continued until the operation run is complete.
(b) If failure occurs between 25 and 100 percent of the endurance run, the test shall be restarted from that particular operation where the failure occurred, and continued until the operation run is complete.
(c) Repeated failure of the same parts, similar parts performing the same function, or similar parts performing dissimilar functions, shall be cause to discontinue the test to correct the deficiency. If the test is discontinued, action shall be taken to determine the cause of failure and correct the deficiency. Verification of the corrective action shall be made to determine the cause of failure and to correct the deficiency. Verification of the corrective action shall be accomplished by a continuation of test in which the test item shall accumulate at least 50 percent of operations without recurrence of the failure.

4.8.2.1.1 Termination of endurance run. In the event that an endurance run is incomplete after 150 percent of operations, or more than three failures have occurred during the test, the test shall not be continued unless approved by NAVSEA. Complete records shall be maintained on each starter test. Each record shall include consolidated tabulation of all previous test problems and failures. If a failure occurs during the run, the cause shall be determined and correction made.

4.8.3 Overload protection. Overload relays shall be tested for tripping characteristics (see 4.8.3.1). Resetting time and temperature compensation shall be as specified in 3.26.2.

4.8.3.1 Tripping characteristics. Overload relays shall be tested in a 50°C ambient at 110, 125, 150, 300, and 500 percent of rated current to determine that the tripping characteristic falls within the acceptable band shown on figure 1. Tests may be conducted at room ambient temperature if the appropriate temperature compensation correction factor for the deviation from 50°C ambient temperature is applied as specified in 3.26.2. Relays shall be tested using the lowest, the highest, and a median rated heater of coil.

4.8.3.2 Special temperature sensing circuitry. Special circuitry for direct temperature sensing and protection of the controlled device shall be tested throughout its operating range with a sensor similar to that used in the protected device.

4.8.4 Vibration. Vibration tests shall be conducted to determine conformance to 3.26.5. Where sway braces must be employed for stability under vibration, the braces shall be constructed to withstand a load five times the



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weight of the unit. This load shall be assumed to be acting at the center-of-gravity of the unit. Any mechanical damage, contact chatter, or electrical malfunction shall constitute a failure.

4.8.5 Shock. Shock tests shall be conducted to determine conformance to 3.26.4. Momentary contact switches and associated switching devices shall be tested in the normal position (not actuated). Other switches that may remain in more than one position shall be tested with contacts actuated and not actuated. The tested controller shall be a completely assembled controller with a load start and stop pushbutton and LVP for each size and application.

4.8.5.1 Test features. Test features shall be as follows:

(a) Type A (for principal units) test shall be performed under light, medium, or heavy weight classifications, as required. Starter and electrically operated components when tested individually shall be shock tested in the motor energized and motor de-energized conditions. When tested separately from the starter, a contactor or relay shall have its operating coil connected in series with its normally open auxiliary contacts in a circuit supplied by rated coil voltage. The starter shall be energized at full load. Specific electrical tests (insulation resistance, dielectric and pick-up and drop-out) shall be made as necessary to determine satisfactory performance following the shock test.

(b) For an assembled starter mounted separately on the ships structure, fixture 4A mounting of MIL-S-901 shall be used. For individual starter components, fixture 6E of MIL-S-901 shall be used. Equipment shall be mounted on the shock machine in a manner simulating normal shipboard mounting.

(c) The following are definitions of failure:

(1) Breaking or cracking of any part, including mounting bolts, or material around the mounting holes that will affect device function.
(2) Failure to pass insulation resistance and pick-up and drop-out tests.
(3) The dielectric strength shall be conducted at a voltage equal to 65 percent of the voltage specified in 3.26.7. Failure to pass this test shall be cause for rejection.
(4) The starter shall be carefully examined to ascertain any mechanical damage. If necessary, the starter shall be partially disassembled to detect any damage. Failure to pass the visual examination shall be cause for rejection.
(5) Appreciable distortion of any parts, including enclosure and framework.
(6) Failure to meet the shock requirements specified in 3.26.4.
(7) Any change of state of the starter that would cause application of power to the motor if the motor was not in operation.
(8) Any change of state of the starter that would require restarting of the motor if the motor was operating.

4.8.5.2 Monitoring of contacts. On shock test of components, each contact shall be monitored. Assembled controllers shall have selected contacts (at least



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30 percent of contacts used in the application) monitored, including one line contact and the contacts most likely to fail. A contact shall be considered as monitored when, during the shock test, the contact is part of an energized circuit where malfunction of the contact would cause the monitor circuit to indicate a malfunction. Closed contacts shall be monitored by a means that will determine if a contact momentarily opens in accordance with 3.26.4(a). Open auxiliary contacts shall be monitored by a means that will determine if a contact momentarily closes. Open line contacts shall be monitored by a means that will indicate if the contacts have momentarily closed during the shock test. Contact monitoring may be accomplished by calibrated relay (except for monitoring open line contacts) oscilloscope, oscillograph, or other appropriate means that is consistently repeatable within plus or minus 10 percent accuracy. At the contractor's option, the contacts may be monitored separately; or, the open contacts may be wired in parallel and monitored as a unit, and the closed contacts wired in series and monitored as a unit, using method 310 of MIL-STD-202.

4.8.6 General operation. The general operation shall include all tests necessary to ascertain that the required operating features function properly. In particular, it shall be determined that the required sequence of operation has been attained and that low-voltage protective devices and accelerating contactors operate as required. Contactors, relays, and other devices having operating coils shall be tested at the specified temperature plus or minus 3°C ambient temperature to determine conformance to 3.16.5, except that these devices may be tested at any convenient ambient temperature between 10 and 40°C in lieu of 65°C for quality conformance inspection.

4.8.7 Inclination. The controller shall be tested in accordance with 3.26.11.

4.8.8 Insulation resistance. The measuring instrument used shall have a full-scale reading not exceeding 200 megohms and an open circuit voltage of 500 Vdc plus or minus 10 percent. Measurement shall be made between all mutually insulated points and between insulated points and ground (see 3.26.6).

4.8.9 Dielectric. Dielectric tests shall be conducted in accordance with NEMA ICS 1 (see 3.26.7). In testing individual limit switches, selector switches, push-buttons, and similar devices with small contact separation distances, a dielectric test shall be made across the contact gap. A test voltage of 1160 Vac applied for 1 second shall be used.

4.8.9.1 Post-shock dielectric. Dielectric tests shall be repeated with 65 percent test voltage.

4.8.10 Effectiveness of enclosure. Enclosures shall be tested to determine conformance to 3.22. Explosion-proof, dripproof, and splashproof enclosure tests are required for qualification inspection only.

4.8.11 Noise, ac contactors and relays (if applicable). Noise tests shall be conducted to determine conformance to the requirements specified in 3.26.16. Ac contactors and relays shall be noise tested in the operating mode.

4.8.12 Pick-up and drop-out voltage. The pick-up and drop-out voltages shall be in accordance with 3.26.14.



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4.8.13 Voltage variation. The controller shall be connected to a voltage supply that can be varied from 80 to 110 percent of the controller rated input voltage to determine conformance to 3.26.1. The controller shall not malfunction due to the input voltage varying between 80 and 110 percent of rated value. The switches shall be operated at the low and high points to determine that the controller operates satisfactorily.

4.8.14 Input harmonic and waveform distortion. The input harmonic and waveform distortion shall be measured as follows:

(a) Operate the starter with the input and output voltage and frequency within the specified steady state range.
(b) Record the input harmonics of the starter up to the fortieth and the deviation factor at no load, 50 percent load, and the rated load.

When tested in this manner, controllers shall meet the requirements specified in 3.26.10.

4.8.15 Efficiency. The starter shall be loaded with resistive load during the test. Meters shall be inserted in the input and output for reading wattage. Readings shall be recorded at no load, half load, and full load at rated input voltage and frequency. Efficiencies shall be calculated at half and full load by means of the following formula:

$$\text{Percent efficiency (overall)} = \frac{\text{Watts (output)} \times 100}{\text{Watts (input)}}$$

These efficiencies shall equal or exceed the manufacturer's rated efficiencies and show conformance to 3.26.9.

4.8.16 Terminal strength. The terminals of the contactors and control relays shall be tested in accordance with test method 211A of MIL-STD-202 (see 3.26.8).

4.8.17 Environmental screening. PLCs shall be subjected to test methods 101, 103, 107, and 110 of MIL-STD-202 (see 3.26.12).

4.8.18 EMI. PLCs shall be subjected to EMI emission and susceptibility tests of MIL-STD-462 to determine conformance to the limits specified in MIL-STD-461 (see 3.26.13).

(a) CE01, CE03.
(b) CS01, CS02, CS06.
(c) RE01, RE02.
(d) RS01, RS02, RS03.

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.


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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.10.)

5.1 Preservation, packing, and marking. When specified (see 6.2), controllers, controller parts, and associated switching devices shall be preserved level A or C, packed level A, B, or C, and marked in accordance with MIL-E-17555.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Controllers, controller parts, and associated switching devices covered by this specification are intended for surface and submarine use. For special duty see 3.7.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Equipment with which associated (for example, pump, motor generator set and so forth).
(d) Quantity required.
(e) Form required (see 1.2.1).
(f) Contactor designation (see 1.2.2).
(g) Associated switching device and controller part designation (see 1.2.3).
(h) If drawings are required (see 3.3).
(i) Duty and service, if special (see 3.7), including when momentary power interruptions or reduced voltage starter recycling permitted by 3.26.4(e) and 4.8.5.1 are not acceptable.
(j) Control voltage, if other than that of the controlled device (see 3.8).
(k) Motor field isolation (see 3.10).
(l) Independently fused indicating and monitoring circuits when required (see 3.11).
(m) Ambient temperature and maximum permissible temperature rise (see 3.12 and table VII).
(n) Ac controllers for low noise, if required (see 3.14 and table VIII).
(o) Emergency-run feature, if required (see 3.16.3).
(p) Local transfer switch for vital function, if required (see 3.16.4).
(q) If embedded temperature detectors are used for overload protection (see 3.16.6).
(r) Ambient pressure compensations, if required (see 3.16.7).
(s) If PLC is to be installed within motor controller enclosure (see 3.16.9).



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(t) If installation footprint for PLC's is other than specified (see 3.16.9.4).
(u) If parts selection is other than specified (see 3.16.9.5).
(v) Input and output type (see 3.16.9.8).
(w) Space heaters, if required (see 3.17).
(x) Indicator lights, if required (see 3.18).
(y) Controller characteristics (see 3.21).
(z) Voltage and phase rating (see 3.21(a)).
(aa) Special service for submarine (see 3.21(a)).
(bb) Current rating required (see 3.21(b)).
(cc) Operation (manual or magnetic), protective features and function (see 3.21(c) and (d)).
(dd) Location and type of switches (see 3.21(e)).
(ee) Dc manual controllers, size, hp, and current, when required (see table IV).
(ff) Motor hp and identification plate amperes (see table V).
(gg) Enclosure type (see 3.22).
(hh) Requirements for inclusion of NSN on identification plate, if needed, and the NSN to be used (see 3.25).
(ii) Provision for wiring to an overspeed trip module, if required (see 3.26.2).
(jj) When the closed line contact opening time under high impact shock conditions is critical (see 3.26.4(a)).
(kk) If timing relays in reduced voltage starters may recycle (see 3.26.4(e)).
(ll) Requirements for resilient mounting of equipment (see 3.26.4.1).
(mm) When controllers are to meet low noise requirements (see 3.26.16).
(nn) Maximum vibration frequency if above 33 Hz (see 3.26.5).
(oo) If inspection conditions are other than specified (see 4.3).
(pp) When the order of tests is other than that specified in 4.7.
(qq) Levels of preservation, packing, and marking required (see 5.1).
(rr) Controller dimensions, if other than that of table VI dimensions.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| 3.3 and appendix | DI-DRPR-80651 | Engineering Drawings | ---- |
| 4.5 | DI-MISC-80678 | Certification data/report | ---- |

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.



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6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. QPL-2212 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity.

6.6 Sampling for quality conformance (see 4.5). Samples should be selected in accordance with MIL-STD-105, general inspection level II, Acceptable Quality Level (AQL) 0.25 single normal.

6.7 Definitions. The definitions specified in 6.7.1 through 6.7.94 apply to terms associated with this specification.

6.7.1 Across-the-line. Across-the-line signifies the connection of an apparatus directly across the main supply circuit.

6.7.2 Ambient temperature. Ambient temperature is the temperature of the medium surrounding an equipment.

6.7.3 Associated switching device. Associated switching devices are devices that are used in conjunction with a controller, but are discrete items which are mounted separately from the controller. These include limit switches, pushbutton stations, pressure switches, and others.

(a) Automatic. Automatic means self-acting, operating by its own mechanism when actuated by some impersonal influence as, for example, a change in current strength, pressure, temperature or mechanical configuration.
(b) Contactor. A contactor is a device for repeatedly establishing and interrupting an electric power circuit.
(c) Contactor, magnetic. A magnetic contactor is a contactor actuated by electromagnetic means.
(d) Contacts. Contacts are conducting parts which co-act to complete or to interrupt a circuit.
(e) Contacts, auxiliary. Auxiliary contacts of a switching device are contacts in addition to the main circuit contacts and function with the movement of the letter.



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6.7.4 Contacts, line. Line contacts are contacts on a contactor or relay used solely for controlling the power to a motor or controlled device.

6.7.5 Contacts, normally open (NO) and normally closed (NC). Normally open and normally closed signify the position taken by contacts when the device is de-energized. The de-energized condition of a manual device is the off position.

6.7.6 Control, local. Local control or device is a control function or device which provides for initiation or change of a control function from within or directly upon the controller enclosure.

6.7.7 Control, remote. Remote control or device is a control function or device which provides for initiation or change of a control function from a remote point.

6.7.8 Control, three-wire. Three-wire control is a control function which utilizes a momentary contact pilot device and a holding circuit contact to provide undervoltage protection.

6.7.9 Control, two-wire. Two-wire control is a control function which utilizes a maintained contact pilot device to provide undervoltage release.

6.7.10 Control circuit. The control circuit of a control apparatus or system is the circuit which carries the electrical signals directing the performance of the controller but does not carry the main circuit power.

6.7.11 Control relay. A control relay is a device for repeatedly establishing and interrupting an electric circuit. A control relay is usually employed to actuate other relays or contactors in the same or associated circuits.

6.7.12 Control-circuit transformer. A control-circuit transformer is a voltage transformer used to supply a voltage for operation of control devices.

6.7.13 Controller, electric motor. An electric motor controller is a device or group of devices which governs, in some predetermined manner, the electric power to the motor. An electric motor controller is distinct functionally from a simple disconnecting device whose principal purpose in a motor circuit is to disconnect the circuit, together with the motor and its controller, from the power source.

6.7.14 Controller, full-magnetic. A full magnetic controller is an electric controller having all of its basic functions performed by devices which are operated by electromagnets. Basic functions usually refers to acceleration, retardation, line closing, reversing, and so forth.

6.7.15 Corrosion-resistant. Corrosion-resistance means constructed, protected, or treated so that corrosion will not exceed specified limits under specified test conditions.

6.7.16 Controller, manual. A manual controller is an electric controller having all of its basic functions performed by hand operated devices.

6.7.17 Disconnection. Disconnection is the opening of a sufficient number



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of conductors to prevent current flow.

6.7.18 Dripproof. Dripproof means constructed or protected so that falling dirt or drops of liquid will not interfere with the successful operation of the apparatus under specified test conditions.

6.7.19 Drop-out, voltage or current. The drop-out voltage or current of a magnetically operated device is the voltage or current at which the device will return to its de-energized position.

6.7.20 Duty, continuous. Continuous duty is a requirement of service which demands operation at a substantially constant load for an indefinitely long time.

6.7.21 Duty, intermittent. Intermittent duty is a requirement of service which demands operation for alternate intervals of (1) load and no-load, or (2) load and rest, or (3) load, no-load and rest; with such alternate intervals being definitely specified.

6.7.22 Duty, varying. Varying duty is a requirement of service which demands operation at wide variations of loads and intervals of time.

6.7.23 Duty of a controller. The duty of a controller is the specific function or functions for which it is constructed with respect to the operation of the motor. These include starting, speed control, reversing, stopping, and the frequency and length of time of operation.

6.7.24 Emergency run. Emergency run is a condition in which the protective devices that normally interrupt a control circuit are temporarily and manually overridden by some means, and cause the circuit function to be maintained.

6.7.25 Enclosure. An enclosure is a surrounding case designed to protect:

(a) Personnel against accidental contact with the enclosed electrical devices, and
(b) Internal devices against specified external conditions.

6.7.26 Equipment. Equipment, as used herein, means controllers, controller parts, and associated switching devices.

6.7.27 Failure of a control component or system. Failure is a state or condition in which a control component or system does not perform its essential functions when its ratings are not exceeded.

6.7.28 Features, distinctive. Distinctive features are exclusive elements which are characteristic of a particular design which are not common to other designs.

6.7.29 Fuse. A fuse is an overcurrent protective device with a circuit opening fusible member which is heated and severed by the passage of over-current through it.

6.7.30 Grounded parts. Grounded parts are parts which are connected in an intentionally grounded manner.



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6.7.31 Insulation. The temperature limits on which the ratings of electrical apparatus are based are largely determined by the character of the insulation materials used.

(a) Insulation is considered to be impregnated when a substance provides a bond between components of the structure and a degree of filling and surface coverage sufficient to ensure adequate performance under the extremes of temperature, surface contamination (moisture, dirt, and so forth) and mechanical stress expected in service. The impregnant should not flow or deteriorate at operating temperatures so as to seriously affect performance in service.

(b) The electrical and mechanical properties of the insulation should not be impaired by the prolonged application of the limiting insulation temperature permitted for the specific insulation class. The word "impaired" is used here in the sense of causing any change which could disqualify the insulating material for continuously performing its intended function, whether creepage spacing, mechanical support or dielectric barrier action.

6.7.32 Interlock. An interlock is a device actuated by the operation of some other device with which it is directly associated to govern succeeding operations of the same or allied devices. Interlocks may be either electrical or mechanical.

6.7.33 Isolation. Isolation is the opening of all conductors connecting to the power source.

6.7.34 Jog (inch). Jog (inch) is a control function which permits the momentary operation of a drive for the purpose of accomplishing a small movement of the driven machine.

6.7.35 Limit. Limit means that the designated quantity is controlled so that a prescribed boundary condition is not exceeded.

6.7.36 Limit, current. Current limit is a control function which prevents a current from exceeding its prescribed limits. Current limit values should be expressed as a percent of rated load value.

6.7.37 LVP. LVP is the effect of a device which, on the reduction or failure of voltage, causes and maintains the interruption of power to the main circuit until the circuit is restarted manually.

6.7.38 LVR. LVR is the effect of a device which, on the reduction or failure of voltage, causes the interruption of power to the main circuit, but does not prevent the re-establishment of the main circuit on return of voltage.

6.7.39 Low voltage release effect (LVRE). LVRE is the effect of a device which remains unchanged on the reduction or failure of voltage and does not cause an interruption of power.

6.7.40 Noise. Noise is an unwanted signal.



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6.7.41 Nonautomatic. Nonautomatic means that the implied action requires personal intervention for its control. As applied to electrical controllers, nonautomatic does not necessarily imply a manual controller, but only that personal intervention is required.

6.7.42 Nonreversing. Nonreversing is a control function which provides for operation in one direction only.

6.7.43 Nonventilated. Nonventilated means constructed to provide no intentional circulation of external air through the enclosure.

6.7.44 Open momentarily, or momentary opening. Where applied in this specification, these expressions mean that an interval of contact physical opening may occur during high impact shock.

6.7.45 Part. A part is an item provided as an integral part of a controller such as a contactor, relay, indicating light, or push-button.

6.7.46 Pick-up and seal voltage. The pick-up and seal voltage of a magnetically operated device is the minimum voltage at which the device moves from its de-energized into its fully energized position.

6.7.47 Pick-up voltage or current. The pick-up voltage or current of a magnetically operated device is the voltage or current at which the device starts to operate.

6.7.48 Protection, overload. Overload protection is the effect of a device operative on excessive current, but not necessarily on short circuit, to cause and maintain the interruption of current flow to the device governed.

6.7.49 Push-button. A push-button switch (push-button) is a master switch having a manually operable plunger or button for actuating the switch.

6.7.50 Push-button station. A push-button station is a unit assembly of one or more externally operable push-button switches, sometimes including other pilot devices such as indicating lights or selector switches, in an enclosure.

6.7.51 Rating, continuous. Continuous rating is the rating which defines the substantially constant load which can be carried for an indefinitely long time.

6.7.52 Rating, 8-hour (as applied to a magnetic contactor). The 8-hour rating of a magnetic contactor is the rating based on its ampere carrying capacity for 8 hours, starting with new clean contact surfaces, under conditions of free ventilation, with full rated voltage on the operating coil, and without causing any of the established limitations to be exceeded.

6.7.53 Rating of a controller. The rating of a controller is an arbitrary designation of an operating limit. It is based on the power governed and on the duty and service required. A rating is arbitrary in the sense that it must necessarily be established by definite fixed standards and cannot, therefore, indicate the safe operating limit under all conditions which may occur.

6.7.54 Reduced voltage. Reduced voltage is the characteristic of a



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controller where the full line voltage is not immediately impressed upon the apparatus, but is applied gradually or in stages by means such as resistors or autotransformers.

6.7.55 Relay, magnetic control. A magnetic control relay is a relay which is actuated by electromagnetic means. When not otherwise qualified, the term refers to a relay intended to be operated by the opening and closing of its coil circuit, and having contacts designed for energizing and de-energizing the coils of magnetic contactors or other magnetically operated devices.

6.7.56 Relay, magnetic overload. A magnetic overload relay is an overcurrent relay in which the electrical contacts are actuated by the electromagnetic force produced by the full or partial load current.

6.7.57 Relay, overload. An overload relay is an overcurrent relay which functions at a predetermined value of overcurrent to cause disconnection of the load from the power supply. An overload relay is intended to protect the load (for example, motor armature) of its controller, and does not necessarily protect itself.

6.7.58 Reset. To reset means to restore a mechanism, storage, or device, to a prescribed state.

6.7.59 Reset, automatic. Automatic reset is a function which operates to automatically reestablish specific conditions.

6.7.60 Reset, manual. Manual reset is a function which requires a manual operation to reestablish specific conditions.

6.7.61 Resistance. Resistance is the (scalar) property of an electric circuit, or of any body, which may be used as part of an electric circuit. Resistance determines, for a given current, the rate at which electric energy is converted into heat or radiant energy, and has a value such that the product of the resistance and the square of the current gives the rate of conversion of energy. In general, resistance is a function of the current, but the term is most commonly used in connection with circuits where the resistance is independent of the current.

6.7.62 Resistance starting. Resistance starting is a form of reduced voltage starting employing resistances which are short-circuited in one or more steps to complete the starting cycle.

6.7.63 Resistant (used as a suffix). Resistant, used as a suffix, means constructed, protected, or treated so that the apparatus will not be damaged when subjected to the specified material or conditions for a specified time.

6.7.64 Resistor. A resistor is a device used to introduce resistance into an electric circuit.

6.7.65 Reversing. Reversing is a control function which permits changing operation of the drive from one direction to the other.



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6.7.66 Rheostat. A rheostat is an adjustable resistor constructed so that its resistance may be changed without opening the circuit in which it is connected.

6.7.67 Service of a controller. The service of a controller is the specific application in which the controller is to be used, for example:

(a) General purpose.
(b) Definite purpose.

(1) Crane and hoist.
(2) Elevator.
(3) Machine tool.

6.7.68 Starter. A starter is an electric controller for accelerating a motor from rest to normal speed. A starter is a device designed for starting a motor in either direction of rotation. If it includes the additional function of reversing, it should be designated a controller.

6.7.69 Starter, autotransformer. An autotransformer starter is a starter provided with an autotransformer which furnishes a reduced voltage for starting. It includes the necessary switching mechanism and is frequently called a compensator or autostarter.

6.7.70 Starter, part-winding. A part-winding starter is a starter which applies voltage successively to the partial; sections of the primary winding of an ac motor.

6.7.71 Starter, primary reactor. A primary reactor starter is a starter which includes a reactor connected in series with the primary winding of an induction motor to furnish reduced voltage for starting. It includes the necessary switching mechanism for cutting out the reactor and connecting the motor to the line.

6.7.72 Starter, primary resistor. A primary resistor starter is a starter which includes a resistor connected in series with the primary winding of an induction motor to furnish reduced voltage for starting. It includes the necessary switching mechanism for cutting out the resistor and connecting the motor to the line.

6.7.73 Submersible. Submersible means constructed to exclude water when submerged in water under specified test conditions of pressure and time.

6.7.74 Switch. A switch is a device for making, breaking, or changing the connections in an electric circuit.

6.7.75 Switch, cam-operated. A cam-operated switch is a switch in which the electrical contacts are opened and closed by a mechanical action of a cam or cams.

6.7.76 Switch, control cutout. A control cutout switch is a switch which interrupts and isolates the control circuit of an electric controller.

6.7.77 Switch, control-circuit limit. A control-circuit limit switch is a



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limit switch in which the contacts are connected only into the control circuit.

6.7.78 Switch, drum. A drum switch is a switch in which the electric contacts are made on segments or surfaces on the periphery of a rotating cylinder or sector, or by the operation of a rotating cam.

6.7.79 Switch, float. A float switch is a switch which is operated by a buoyant constituent part and is responsive to the level of liquid.

6.7.80 Switch, limit. A limit switch is a switch which is operated by some part or motion of a power-driven machine or equipment to alter the electrical circuit associated with the machine or equipment.

6.7.81 Switch, master (see also push-button). A master switch is a switch which dominates the operation of contactors, relays, or other remotely operated devices.

6.7.82 Switch, pressure. A pressure switch is a switch which is operated by a constituent part and is responsive to fluid (gas or liquid) pressure.

6.7.83 Switch, proximity. A proximity switch is a device which reacts to the proximity of an actuating means without physical contact or connection therewith.

6.7.84 Switch, selector. A selector switch is a manually operated, multiposition switch for selecting an alternative control circuit.

6.7.85 Temperature, ambient. Ambient temperature is the temperature of the medium such as air, water, or earth into which the heat of the equipment is dissipated. For self-ventilated equipment, the ambient temperature is the average temperature of the air in the immediate neighborhood of the equipment. For air- or gas-cooled equipment with forced ventilation or secondary watercooling, the ambient temperature is taken as that of the in-going air or cooling gas. For self-ventilated enclosed (including oil-immersed) equipment considered as a complete unit, the ambient temperature is the average temperature of the air outside of the enclosure in the immediate neighborhood of the equipment.

6.7.86 Temperature rise. Temperature rise refers to the increase in temperature above ambient, measured in °C, that will be reached by a device during its normal operation.

6.7.87 Tests, dielectric. Dielectric tests are tests which consist of the application of a voltage higher than the rated voltage for a specified time for the purpose of determining the adequacy against breakdown of insulating materials and spacings under normal conditions.

6.7.88 Thermal cutout. A thermal cutout is an overcurrent protective device which contains a heater element in addition to and affecting a fusible member which opens the circuit.

6.7.89 Thermal protector (as applied to motors and generators). A thermal protector is a protective device which is intended for assembly as an integral part of the machine and which, when properly applied, protects the machine against dangerous overheating due to overload and, in a motor, failure to start.



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6.7.90 Time delay. Time delay means that a time interval is purposely introduced in the performance of a function.

6.7.91 Transition, closed-circuit. Closed-circuit transition, as applied to reduced-voltage controllers, including star delta controllers, is a method of starting in which the power to the motor is not interrupted during the starting sequence.

6.7.92 Ventilated. Ventilated means constructed to provide for the circulation of external air through the enclosure to remove heat, fumes, or vapors.

6.7.93 Watertight. Watertight means constructed to exclude water applied in the form of a hose stream under specified test conditions.

6.7.94 Waveform, output. Output waveform is the graphical representation of the output during one pulse interval.

6.8 Special information concerning shockproofness. Users of this specification who design equipment or acquire controllers and associated switching devices should be aware of the extent of shockproofness of equipment listed on the Qualified Products List QPL-2212. It is the aim of this specification to provide equipment as shockproof as possible for Naval shipboard use; however, due to the nature of shock, 100 percent shockproofness is not possible in all cases. During shock, it is possible that the controller output power may be interrupted momentarily. It is also possible that associated switching devices may interrupt the circuit they are part of momentarily.

6.8.1 Limits on shock testing. Users of this specification who have the responsibility for acquisitioning shockproof equipment for Naval shipboard use are requested to give careful consideration before requiring additional shock testing, which will stress the equipment and may cause unnoticed damage. If additional shock testing is deemed essential, then instruction should be issued to require careful post-shock examination by qualified personnel.

6.9 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.9.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

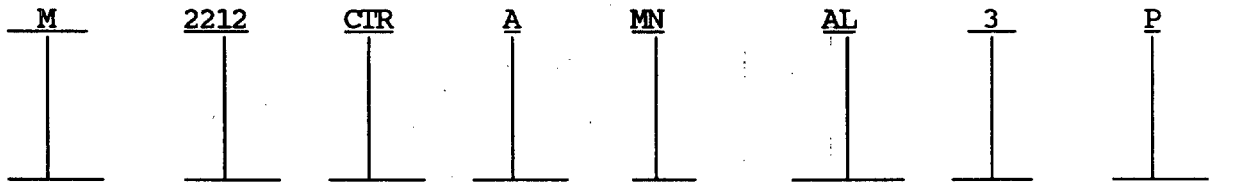
6.10 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.11 Part or Identifying Number (PIN) structure. The PIN's should be constructed as shown below.



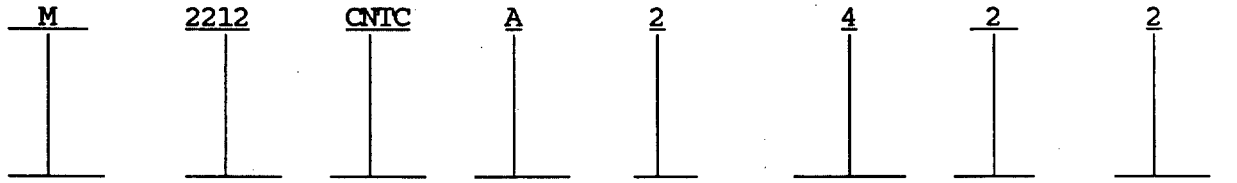
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6.11.1 Controller.




| Prefix to indicate an item defined by inch-pound units. | Specification number | Equipment | Power (see 1.2.1.1) | Type (see 1.2.1.2) | Configuration (see 1.2.1.3) | Size (see 1.2.1.4) | Protective feature (see 1.2.1.5) |
|---|---|---|---|---|---|---|---|

6.11.2 Contactor.




| Prefix to indicate an item defined by inch-pound units. | Specification number | Equipment | Current (see 1.2.1.1) | Size (see 1.2.1.4) | Poles (see 1.2.2.1) | NO AUX contacts (see 1.2.2.2) | NC AUX contacts (see 1.2.2.3) |
|---|---|---|---|---|---|---|---|

6.11.3 Associated switching device and controller part.




| Prefix to indicate an item defined by inch-pound units. | Specification number | Equipment type (see 1.2.3.1) | Device or part (see 1.2.3.2) |
|---|---|---|---|



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6.12 Subject term (key word) listing.

Controller, magnetic
Controller, manual
Enclosure, controller
Relay
Switches, limit, pressure, selector, and temperature
Transformers, ac and dc

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6110-N313)



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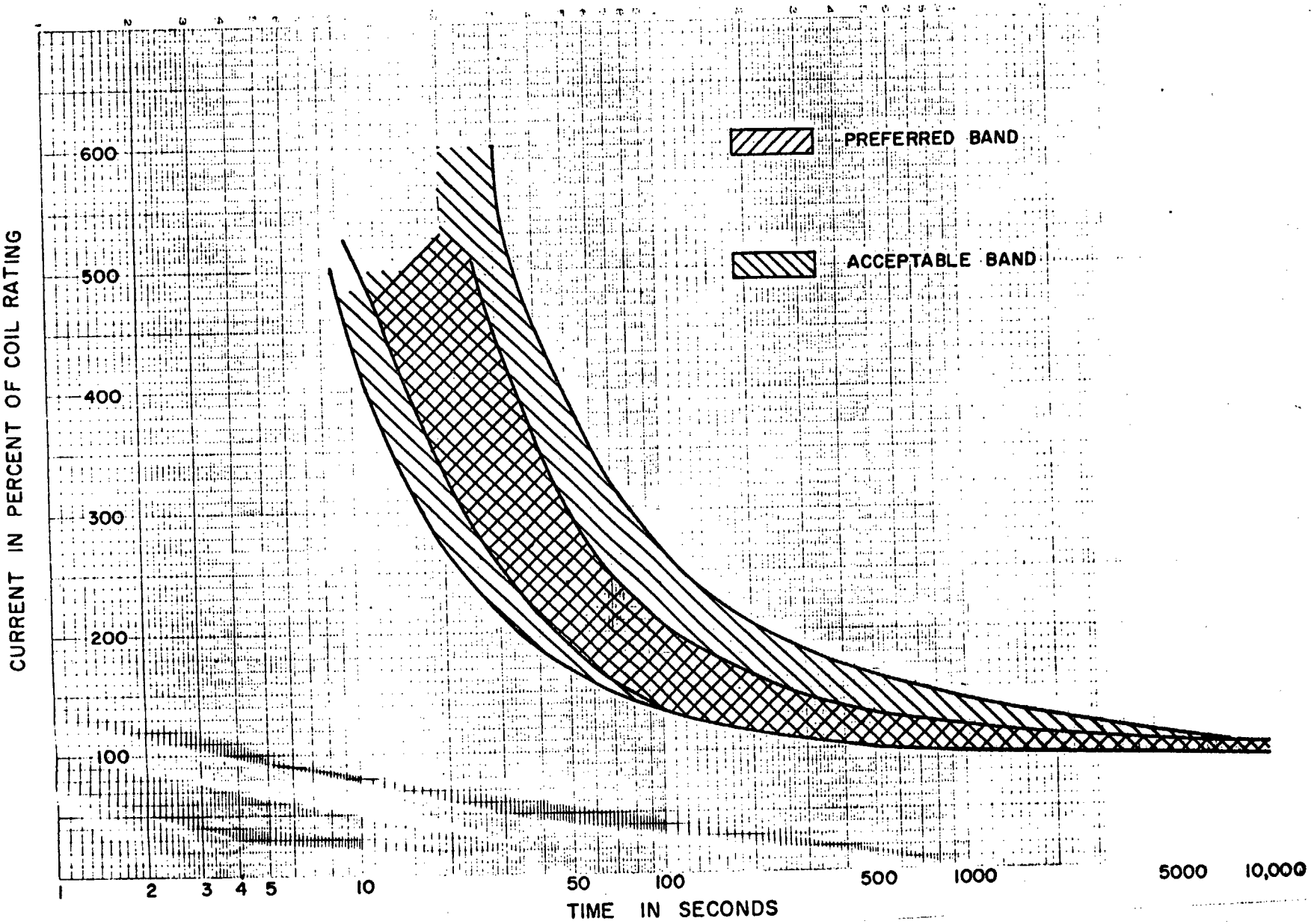



FIGURE 1. Time current characteristic of overload relays.



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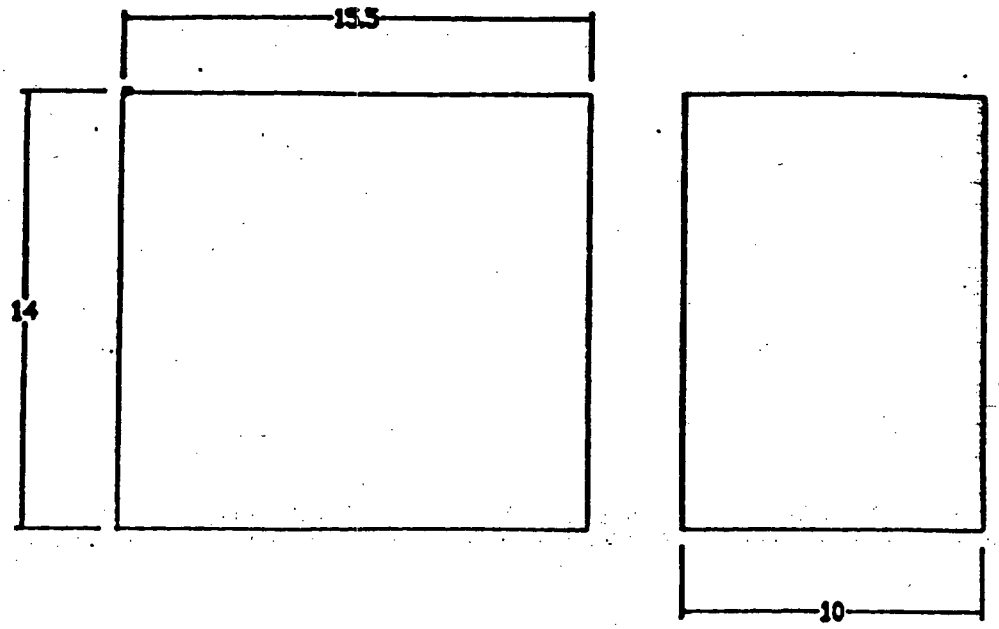



FIGURE 2. PLC installation envelope.

NOTE: All dimensions in inches.

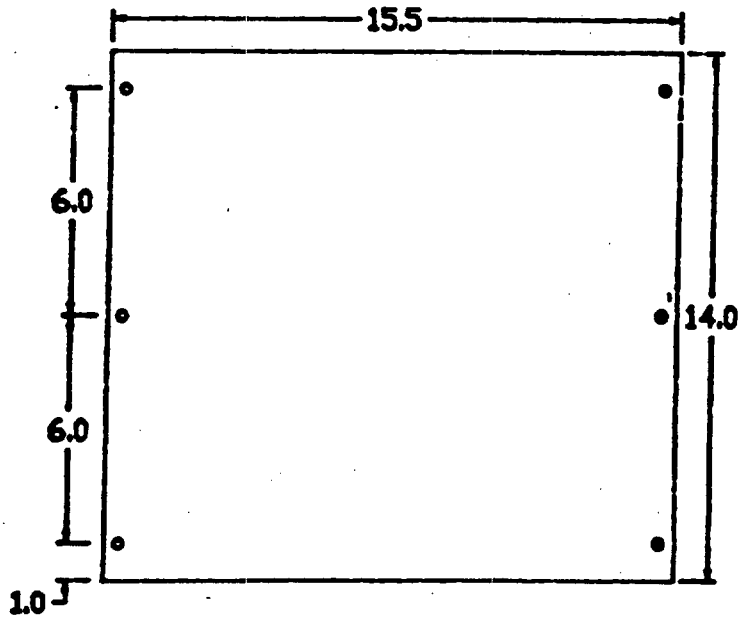



FIGURE 3. PLC mounting footprint.



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# APPENDIX A

## ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

### 10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

### 20. DRAWING CONTENT

20.1 Drawing content. Drawings shall contain the information specified in 20.1.1.

20.1.1 Drawings (general). Drawings for contractors, controllers, and associated switching devices shall contain the following:

a. List of material, listing the enclosures and all parts. Each item shall be identified by the catalog number assigned by the company responsible for the design of the item. Navy standard items shall be identified by the drawing number or NSN.

b. Outline of the front and top views of the controller, giving dimensions and basic construction details such as location of parts, mounting panels, main mounting brackets, and main structural features including channels, gussets, and so forth. Dimensions shall include overall size, location and size of mounting holes, clearance required to open door, location of drain holes, location and size of cable entrance holes, and so forth.

c. Connection and schematic diagrams of the control circuit, including connections to associated switching devices and other remote devices. The rating of all fuses shall be indicated.

d. Description of operation of the controller.

e. Description of adjustments which can be made on the equipment such as the range of time for a timing relay and factory equipment settings that have been made.

f. Table of overload heater coil catalog numbers with motor full-load current ranges.

g. Current transformer data including primary to secondary current ratio and frequency.

h. Primary and secondary voltage and volt-ampere rating of control circuit transformers.



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i. Table of autotransformers of reactor data showing wire size, insulating materials, and insulation processes.

j. Table of applicable resistor catalog numbers with corresponding step-by-step and overall resistances, continuous wattage rating of each step and identification as to where used.

k. Capacitor data, including type and rating.

l. List of exceptions to this specification.

m. Descriptive data of the controller which shall include rating, degree of enclosure, catalog number, and size of designation. The data shall include the shock test reference which will entail the shock test report number, data, and NAVSEA approval letter date.

n. Table of applicable shunt coil catalog numbers with corresponding voltage, ohms, turns, wire size, covering, coil insulation and impregnation and devices on which used.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-2212G(SH) | CONTACTORS AND CONTROLLERS, ELECTRIC MOTOR AC OR DC, AND ASSOCIATED SWITCHING DEVICES. |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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